

1 Orders and lattices

A partially ordered set is the basic object of this section. We begin with the definitions and then prove a few elementary facts.

Definition 1.1. A *partial order* on a set P is a relation $\leq$ that is reflexive, antisymmetric and transitive. The pair $(P, \leq)$ is a *poset*.

Definition 1.2 (Upper bound). Let $A \subseteq P$. An element $u \in P$ is an *upper bound* of A if $a \leq u$ for every $a \in A$. The least upper bound, when it exists, is written $\sup A$.

Example 1. The power set $\mathcal{P}(X)$ ordered by inclusion is a poset in which every family $\mathcal{F}$ has the supremum $\bigcup \mathcal{F}$.

Lemma 1.3. A subset of a poset has at most one least upper bound.

Proof. If u and v are both least upper bounds of A , then $u \leq v$ and $v \leq u$, so $u = v$ by antisymmetry. $\square$

Theorem 1.4 (Knaster–Tarski). Let L be a complete lattice and let $f: L \rightarrow L$ be order preserving. Then the set of fixed points of f is itself a complete lattice; in particular f has a least fixed point.

Remark 1.5. Completeness cannot be dropped: the map $n \mapsto n + 1$ on $\mathbb{N}$ preserves order and has no fixed point.

Corollary 1.6. Every order preserving map on a finite lattice has a fixed point.

2 Chains

Proposition 2.1. Every finite nonempty chain has a greatest element.

Proof. Induct on the size n of the chain. For $n = 1$ there is nothing to show. Otherwise remove any element x , take the greatest element m of the rest, and compare x with m . $\square$

Example 2 (Integers). The chain $(\mathbb{Z}, \leq)$ has no greatest element, so finiteness in Proposition 2.1 is needed.

Conjecture. Every sufficiently regular poset of width w can be covered by w chains that are computed greedily.

Exercise. Show that the divisibility order on $\{1, 2, \dots, 12\}$ is not a chain.

Note. Starred environments are unnumbered and do not advance any counter.

Theorem 2.2. Every poset can be extended to a total order on the same set.